

La ecuación de mov. $\ddot{x} \cosh^2 x + \dot{x}^2 \cosh x \sinh x + g \sinh x = 0$ (0)

H? , Ecs de Hamilton?

Ligadura $y = f(x) \Rightarrow \dot{y} = \frac{df(x)}{dt} = \frac{df(x)}{dx} \frac{dx}{dt} = f' \dot{x}$

$$L = T - V = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - mgy = \frac{1}{2} m \dot{x}^2 (1 + f'^2) - mgf$$

$$\rightarrow L = \frac{1}{2} m \dot{x}^2 (1 + f'^2) - mgf \quad ; * \frac{d}{dt} (\partial_{\dot{x}} L) - \partial_x L = 0$$

$$\partial_{\dot{x}} L = m \dot{x} (1 + f'^2) \rightarrow \frac{d}{dt} (\partial_{\dot{x}} L) = m \ddot{x} (1 + f'^2) + 2 m \dot{x} f' f'' \dot{x} \\ = m \ddot{x} (1 + f'^2) + 2 m \dot{x}^2 f' f''$$

$$\partial_x L = \frac{1}{2} m \dot{x}^2 (2 f' f'') - mg f'$$

Ecuación de mov: $m \ddot{x} (1 + f'^2) + 2 m \dot{x}^2 f' f'' - m \dot{x}^2 f' f'' + mg f' = 0$

$$\rightarrow \ddot{x} (1 + f'^2) + \dot{x}^2 f' f'' + g f' = 0 \quad (\Delta)$$

$$(0) = (\Delta) \quad (1 + f'^2) = \cosh^2 x \quad ; f' f'' = \cosh x \sinh x, f' = \sinh x$$

$$\therefore \text{Si } f' = \sinh x \Rightarrow f(x) = \cosh x$$

$$\therefore L = \frac{1}{2} m \dot{x}^2 (1 + \sinh^2 x) - mg \cosh x \quad ; 1 + \sinh^2 x = \cosh^2 x$$

$$\Rightarrow L = \frac{1}{2} m \dot{x}^2 \cosh^2 x - mg \cosh x \quad ; * p_x = \partial_{\dot{x}} L$$

$$p_x = m \dot{x} \cosh^2 x \quad / \quad \rightarrow p_x^2 = m^2 \dot{x}^2 \cosh^4 x \rightarrow m \dot{x}^2 \cosh^2 x = \frac{p_x^2}{m \cosh^2 x}$$

$$* H = p_x \dot{x} - L = \frac{1}{2} \frac{p_x^2}{m \cosh^2 x} + mg \cosh x \rightarrow \dot{x} = \frac{p_x}{m \cosh^2 x}$$

$$\Rightarrow H = \frac{1}{2} \frac{p_x^2}{m \cosh^2 x} + mg \cosh x \quad ; * \dot{p}_j = -\partial_{q_j} H \quad ; \dot{q}_j = \partial_{p_j} H$$

$$\therefore \dot{x} = \partial_{p_x} H = \frac{p_x}{m \cosh^2 x}$$

$$\dot{p}_x = -\partial H = \frac{p_x \sinh x}{m \cosh^3 x} - mg \sinh x$$

(a) Pruebe que la cte Transformación es canónica

$$Q = \rho \cot \varphi, \quad P = \log\left(\frac{\sin \varphi}{\rho}\right)$$

(b) Encuentre la función generatriz $F_1(\varphi, Q)$ para esta transformación $\star \csc^2 \theta = 1 + \cot^2 \theta$

$$\star \{f, g\} = \sum_{j=1}^n \left[\partial_{q_j} f \partial_{p_j} g - \partial_{p_j} f \partial_{q_j} g \right] \Rightarrow \{Q, P\} = \partial_Q Q \partial_P P - \partial_P Q \partial_Q P$$

$$\partial_Q Q = -\rho \csc^2 \varphi \quad ; \quad \partial_P P = \frac{\rho}{\sin \varphi} \left(\frac{\sin \varphi}{\rho} \right)' = \frac{\rho}{\sin \varphi} \left(-\frac{\sin \varphi}{\rho^2} \right) = -\frac{1}{\rho}$$

$$\partial_P Q = \cot \varphi \quad ; \quad \partial_Q P = \frac{\rho}{\sin \varphi} \left(\frac{\sin \varphi}{\rho} \right)' = \cot \varphi$$

$$\Rightarrow \{Q, P\} = (-\rho \csc^2 \varphi) \left(-\frac{1}{\rho} \right) - \cot^2 \varphi = \csc^2 \varphi - \cot^2 \varphi$$

$$\Rightarrow \boxed{Q \text{ y } P \text{ son canónicas}} ; \star \int \tan x dx = \int \frac{-\ln(\cos x) + C}{\ln(\sec x) + C} ; \int \ln x dx = x \ln x - x + C$$

Función generatriz $F_1(\varphi, Q)$ (tipo 1) se tienen las relaciones

$$P = \partial_Q F_1, \quad \varphi = -\partial_Q F_1$$

Despejar ρ de la transformación en términos de (φ, Q)

$$Q = \rho \cot \varphi \Rightarrow \rho = Q \tan \varphi = \partial_Q F_1 \Rightarrow F_1(\varphi, Q) = Q \int \tan \varphi d\varphi$$

$$\Rightarrow F_1(\varphi, Q) = -Q \ln(\cos \varphi) + g(Q)$$

$$\text{Ahora, con } P = -\partial_Q F_1 = \ln(\cos \varphi) - g'(Q) \quad (\bullet) \quad P = \log\left(\frac{\sin \varphi}{\rho}\right)$$

$$P = \log\left(\frac{\sin \varphi}{Q \tan \varphi}\right) = \log\left(\frac{\sin \varphi}{Q \frac{\sin \varphi}{\cos \varphi}}\right) = \log\left(\frac{\cos \varphi}{Q}\right) = \log(\cos \varphi) - \ln Q \quad (\blacktriangle)$$

$$(\bullet) = (\blacktriangle) \quad \ln(\cos \varphi) - g'(Q) = \ln(\cos \varphi) - \ln Q \quad ; \quad \log x = \ln x \quad (\text{base } e)$$

$$\Rightarrow g'(Q) = \ln Q \Rightarrow g(Q) = \int \ln Q dQ = Q \ln Q - Q + C$$

(por conveniencia $C=0$, no afecta generación canónica, P y ρ no cambian)

$$\therefore \boxed{F_1(\varphi, Q) = -Q \ln(\cos \varphi) + Q \ln Q - Q}$$